Industry Heterogeneity, LLM Release Processes, and Forecasts of Small-Business AI Use

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Abstract

We examine industry differences in small-business artificial intelligence (AI) use and evaluate forecasts of published adoption rates three Census releases ahead. A reconstruction of the Business Trends and Outlook Survey reproduces the original Milestone 2 baseline. The extended analysis compares industry models, partial pooling, hierarchical subset selection with nonlinear terms, regularized regression, and gradient boosting under nested temporal validation. The evaluation comprises 706 forecasts at 10 issue dates. Industry models achieve the lowest aggregate MAE, 3.04 percentage points, compared with 3.39 for the six-release mean and 3.30 for the baseline. Nonlinear terms are repeatedly selected, but their subset model does not improve aggregate accuracy over the baseline. Industry growth estimates are accompanied by uncertainty intervals and rank distributions. A controlled random-feature experiment evaluates prediction beyond the interpolation threshold. A benchmark-marked nonhomogeneous Poisson process models LLM releases, with separate regressions for arrival intensity and capability marks. Its forward forecast is 12.48 benchmarked-catalog releases over 42 days. Benchmark and arrival features provide no clear incremental adoption-forecast improvement over matched controls. Retrospective benchmark calibration, selective catalog coverage, the short survey window, and clustered releases restrict inference.

1 Introduction

Small-business support organizations may use forecasts of AI adoption to anticipate demand for information, training, and implementation assistance. Such forecasts require evidence that improvements persist across industries, business sizes, and forecast dates. Differences in observed adoption also require uncertainty assessment: industries with similar estimated growth rates may have substantially overlapping rank distributions.

This study addresses four questions. First, do industry-specific relationships improve forecasts beyond a pooled linear baseline? Second, do quadratic terms and interactions provide predictive information after temporal validation? Third, how does forecast error change as the number of coefficients approaches and exceeds the number of training rows? Fourth, can a benchmark-marked release process, with a separately estimated nonstationary arrival rate, supply useful predictors of AI adoption? The response is an industry-by-employment-size group's published AI-use percentage, rather than a firm-level adoption indicator or a measure of outreach effectiveness.

2 Data and study population

2.1 Source and reconstruction

The Census Business Trends and Outlook Survey (BTOS) publishes estimates for employer businesses. We restrict attention to five employment-size categories: 1–4, 5–9, 10–19, 20–49, and 50–99 employees. The comparable series begins with collection on November 17, 2025, when the AI question was broadened to cover any business function. Earlier observations are excluded because the wording change introduced a discontinuity [2].

The original workbook was absent from the supplied project, and the Census download host returned timeouts during acquisition. The analysis therefore uses the unmodified, attributed CSV redistribution by The Uptime Report, which records a Census workbook extraction on October 15, 2026 [3]. The data include estimates, standard errors, collection dates, and national, industry, and size aggregates through release 202620. The file hash and acquisition record are supplied in the repository. This source reproduces the original report's historical cell counts and forecasting results; byte identity with the missing workbook cannot be established. Other business-condition survey questions are unavailable in the extract and do not enter the reported fits.

The original historical window contains 22 releases through 202619: 2,200 potential cells and 498 unavailable current-use estimates. The reconstructed analysis adds release 202620, giving 2300 potential cells across 23 releases. Missing rows in the extract are treated as unavailable values. Their individual original suppression flags cannot be recovered. BTOS withholds estimates for quality or disclosure concerns, including relative standard errors above 50% [1]. Missing current-use rates and forecast outcomes are never imputed.

2.2 Screening and temporal alignment

A group is retained when both current and expected AI use are published in at least seven of the first eight releases. Multi-sector category XX is excluded. The rule retains 71 groups in 15 industries and is applied before the first evaluation date. The resulting population consists of relatively well-measured industry-size groups. It does not represent all small businesses, and group-level averages are not business-population-weighted averages.

Let $y_{g,t}$ denote the current-use percentage for group g at release t, and let $e_{g,t}$ denote expected use over the next six months. We predict $y_{g,t+3}$ at the publication of release t. Three releases correspond to approximately six weeks. Reference periods, collection periods, and publication dates are distinct. For the included extract, publication is reconstructed as the Thursday following collection; this schedule reproduces the original report's forecast dates. Event analyses use the previous fortnight's reference period, with its end one day before collection begins. The calendar is exported for inspection.

There are 1406 complete issue–target pairs in the extended sample. Historical evaluation covers the original nine issue dates; the extension contains one additional issue date. All expanded specifications were developed after the extension outcome was publicly available. Consequently, the extension is a retrospective temporal test of these specifications, rather than a newly conducted prospective experiment.

3 Statistical methods

3.1 Forecast models

The original baseline is pooled ordinary least squares (OLS):

$$y_{g,t+3} = \beta_0 + \beta_y y_{g,t} + \alpha_{s(g)} + \delta_{b(g)} + \beta_t t + \varepsilon_{g,t+3}, \tag{1}$$

where $s(g)$ and $b(g)$ identify industry and size. Benchmarks carry forward the latest rate or average the latest six available releases. Separate industry regressions allow each industry's coefficients to differ. A partial-pooling estimator retains unpenalized global intercept, current-use, and time coefficients, while ridge penalties shrink industry and size intercept deviations and industry trend deviations. This estimator is a penalized regression; it is not a fitted random-effects likelihood.

The full feature dictionary includes current and expected use, linear time, squared current use, squared expected use, squared time, current-use-by-time and current-by-expected-use products, industry-specific current-use slopes and trends, size-specific trends, lags of 1, 2, 3, and 6 releases, a six-release average, recent change, the expectation gap, the current estimate's standard error, and available national, industry, and size AI-use and expectation aggregates. It also includes a known sample-year indicator and sine/cosine terms with period six releases. The latter investigate the rotating-panel cycle; the available history cannot establish annual seasonality. Supplementary missing predictors are median-imputed using training rows only, with missingness indicators. Numerical features are standardized within each training fold.

We compare full OLS, ridge regression, elastic net, and histogram gradient boosting. Ridge penalties are selected from $\{0.1, 1, 10, 100, 1000, 10000, 100000\}$. Elastic net uses a mixing fraction of 0.5 and penalties $\{0.01, 0.1, 0.5, 1\}$. Boosting uses 150 iterations, learning rate 0.05, at least 25 observations per leaf, an L_2 penalty of 10, and maximum leaf counts $\{3, 7, 15\}$. These finite dictionaries constrain the search in a short panel.

3.2 Hierarchical subset selection and nonlinearity

Subset selection retains the baseline's main effects and exhaustively searches optional blocks. A linear comparison permits expected use and industry- or size-specific slopes and trends. The nonlinear search additionally permits y^2, e^2, t^2, yt, and ye. Industry indicators and their interactions enter as blocks. Strong hierarchy requires the expected-use main effect whenever e^2 or ye is selected; the remaining parent effects are already mandatory. Each of the 320 admissible nonlinear candidates is fitted by least squares. Selection minimizes average validation-origin MAE; exact ties favor fewer columns. This is best subset selection over a specified block dictionary, rather than exhaustive selection over every derived predictor [4].

A quadratic term or interaction remains linear in its fitted coefficient while allowing nonlinear or industry-dependent response relationships. Its selection is evidence of predictive utility within the candidate dictionary. Selection frequencies across outer origins measure stability. Conventional coefficient tests performed after searching would omit selection uncertainty, so we do not interpret selected-term p-values as confirmatory evidence.

3.3 Nested temporal evaluation

At an outer issue date d_t, the training sample satisfies

$$\mathcal{T}_t = \{(g, u) : \text{publication}(u + 3) \leq d_t\}. \tag{2}$$

The outer test consists of complete pairs issued at t. Within $\mathcal{T}_t$, up to four earlier validation origins span the available history. Each inner fit again excludes targets unpublished at its own issue date. Subset choice, regularization, and boosting complexity use only these inner predictions. Screening, preprocessing, and selection therefore do not use outer outcomes. Historical data vintages cannot be reconstructed from the redistributed file, so this is temporal validation of the recorded vintage rather than a complete real-time vintage experiment.

Forecast accuracy is evaluated by mean absolute error (MAE), root mean squared error (RMSE), and bias, all in percentage points (pp), with industry and size breakdowns. Predictions are not clipped to $[0, 100]$; the frequency of violations is recorded. Paired differences in origin-level MAE are summarized with 2,000 circular moving-block bootstrap draws of length three origins. These intervals address dependence across nearby forecasts approximately. With only 10 origins they remain exploratory. Exact sign-flip calculations are supplied as a sensitivity calculation under an exchangeability assumption; they are not relied upon for inference under serial dependence. Comparisons across several models introduce additional selection uncertainty.

Survey standard errors quantify uncertainty in the published target. We report their relationship to forecast errors without treating them as a strict lower bound on MAE. Sampling error, temporal dependence, model error, and uncertainty about the underlying adoption rate are distinct quantities.

3.4 Industry growth and ranking uncertainty

For descriptive industry growth, we fit

$$y_{g,t} = a_g + \gamma_{s(g)}(t/2) + u_{g,t}, \tag{3}$$

with group intercepts a_g. The coefficient γ_s is the change in percentage points per four weeks, using two releases per four weeks. Equal observation weights define the primary estimand; retained size bands need not be identical across industries.

Uncertainty is estimated with 2,000 wild residual bootstrap draws. Six-release blocks share multipliers across all groups, and block boundaries have random offsets. This retains common release shocks and approximates within-cycle dependence. We report percentile intervals, median ranks, rank ranges, and the proportion of bootstrap draws placing an industry in the top three. These proportions describe resampling stability, not posterior probabilities. The small number of temporal blocks and residual-model assumptions restrict precision. Pairwise difference intervals are exploratory and are not simultaneous confidence intervals. Sensitivity fits add a common indicator at release 202616, when the annual sample changes, or weight observations inversely by the squared published standard error with a 1-pp standard-error floor.

3.5 Model-release timing

Six provider announcements form a selected chronology: Gemini 3, Claude Opus 4.5, GPT-5.2, Claude Opus 4.6, Gemini 3.1 Pro, and GPT-5.4 [6, 7, 8, 9, 10, 11]. For each event, we compare three complete reference fortnights before the announcement with three wholly subsequent fortnights, excluding a straddling window. Events without three pre- and post-event observations are marked unsupported. A constant set of groups with observed rates at every release prevents changing suppression patterns from mechanically shifting the aggregate.

We also regress the balanced-group mean on time, a sample-change indicator, and the cumulative number of selected announcements. Releases overlap in time and exposure is common to all industries. The resulting coefficient is descriptive: adoption trends, survey changes, and

other contemporaneous events supply alternative explanations. The design does not identify an announcement's causal effect.

3.6 Overparameterization experiment

To vary parameter count while holding input information fixed, we transform standardized baseline and expected-use inputs into nested random Fourier features,

$$z_j(x) = \sqrt{2/m}\cos(\omega_j^\top x + b_j), \quad \omega_j \sim N(0, I/d), \quad b_j \sim U(0, 2\pi), \tag{4}$$

where d is input dimension. Feature centering uses training rows. Three seeds and the first, middle, and final outer origins are evaluated at coefficient-to-row ratios near 0.05, 0.2, 0.5, 0.8, 1, 1.2, and 2. All widths use the same initial feature sequence within a seed. The intercept is included in coefficient counts and matrix rank.

For each width we fit minimum-norm least squares and ridge regression with fixed penalty 0.01. The ridge objective uses the unnormalized residual sum of squares, and the penalty is fixed as a diagnostic rather than tuned on test outcomes. We record rank, effective degrees of freedom, condition number, and training and future errors. Numerical interpolation is assessed directly. Increasing p beyond n does not by itself establish interpolation when the matrix is rank deficient. A fall in future error after a peak near interpolation would be consistent with double descent [5]; its occurrence is evaluated empirically for this feature family.

3.7 Benchmark-marked LLM release data

The release analysis uses Epoch AI's October 30, 2026 catalog, containing 271 canonical model records with release dates and Epoch Capabilities Index (ECI) scores. The intensity regression uses the 221 records released from January 1, 2024 through October 30, 2026. This definition includes model variants that Epoch records separately and excludes releases absent from its scored catalog. The target process is therefore the arrival of benchmarked catalog models, not the complete population of LLM releases. Provider-by-date batches are examined as an alternative event unit.

ECI combines benchmark results through an item-response model and provides a common capability scale [12]. Its numeric origin is a calibration convention. We use ECI as a mark attached to each event and GPQA Diamond as a supplementary, directly interpretable benchmark percentage. For models with multiple GPQA evaluation variants, their median is used. The original files, source URLs, hashes, license attribution, and canonical model mapping are retained.

Historical model release dates are available, but historical score-publication timestamps and benchmark vintages are not. A model released before an issue date can receive an ECI estimate calibrated using later evaluations. Restricting events by release date therefore does not establish that their current scores were available at the time. All benchmark-augmented adoption comparisons are retrospective analyses of this fixed score vintage. No real-time deployment accuracy is claimed.

3.8 Nonhomogeneous Poisson intensity regression

Let $N(A)$ count catalog model releases in a time interval A. Conditional on an intensity function, the nonhomogeneous Poisson process (NHPP) assumes independent increments and

$$N([a,b]) \sim \text{Poisson}(\Lambda(a,b)), \qquad \Lambda(a,b) = \int_a^b \lambda(u)\,du. \tag{5}$$

Dates are observed at daily resolution. We use a piecewise constant daily intensity, so the integrated intensity is an exact sum of daily means. Zero-release days are retained. If Y_d is the number of releases on day d, the rate regression is

$$Y_d \sim \text{Poisson}(\lambda_d), \qquad \log \lambda_d = \beta_0 + \beta_1 t_d + \beta_2 W_d, \tag{6}$$

where t_d is elapsed years since January 1, 2024 and W_d indicates Monday through Friday. The exposure is one day. Poisson GLM maximum likelihood estimates the coefficients [13]; a seven-day HAC covariance supplies approximate coefficient uncertainty. Calendar effects make the rate nonstationary even without a positive secular trend.

The candidate rate regressions are constant intensity, linear time, quadratic time, linear time plus weekday, and linear time plus an annual sine/cosine pair. At each issue date, the specification is selected by mean daily negative log likelihood over six earlier, disjoint 42-day windows. Each validation window has its own fit using only preceding dates and at least one year of history. The final October 30 forecast repeats this procedure. Selection scores are reported as selection criteria, not as an independent test of the chosen final specification.

Pearson dispersion, residual lag-one correlation, and the frequency of days with multiple releases assess the independent-Poisson assumption. A plug-in simulation compares observed multi-release days with 2,000 Poisson replications. Provider-day grouping assesses sensitivity to joint announcements. An additional predictive sensitivity replaces daily Poisson counts with moment-matched negative-binomial counts having mean λ_d and variance $\phi\lambda_d$, where ϕ is the selected model's Pearson dispersion. This sensitivity is an overdispersed count model, rather than an NHPP.

3.9 Regression of benchmark marks and future releases

For a released model j, capability is modeled as

$$C_j = \alpha_0 + \alpha_1 t_j + \eta_j. \tag{7}$$

OLS estimates the conditional mean ECI among catalog releases, rather than the frontier trajectory. Parameter covariance is clustered by provider and release week. Future arrivals are simulated from independent daily Poisson increments, and each receives a regression-predicted ECI plus a sampled fitted residual. The simulated frontier is the maximum of the current observed frontier and the marks of future arrivals.

For the forward 42-day forecast, Gaussian coefficient draws from the estimated rate and mark covariances approximately propagate parameter uncertainty; non-positive numerical covariance eigenvalues are truncated to zero for simulation. These intervals remain conditional on the catalog, regression forms, mark residual distribution, and Poisson assumption. They do not account fully for selective benchmark coverage, future benchmark revisions, dependence among future releases, or changing developer composition. The overdispersed count interval is reported separately. A future model name or exact release date is not inferred.

3.10 Linking the release process to adoption

The known release predictors are the preceding 42-day release count, released-model ECI frontier, mean recent ECI, frontier gain, released-model GPQA frontier, and a decaying capability index,

$$S(d) = \sum_{j:T_j \le d} \max\left\{\frac{C_j - 100}{10}, 0\right\} \exp\left(-\frac{d - T_j}{90}\right). \tag{8}$$

The reference of 100 ECI and decay of 90 days are fixed index conventions. They do not define economically equivalent units of AI capability.

Future predictors are the integrated expected arrival count, expected capability-weighted arrival load, and Monte Carlo expected frontier gain. They are computed from regressions refitted at each row's own issue date. Their endpoint is the target survey's reference-period end. In the included calendar, a $t+3$ outcome is published 42 days after the issue date but refers to a period ending 18 days before that publication; hence the unknown future exposure is 24 days. Releases after that reference end cannot explain the target and are excluded. This exposure horizon is separate from the 42-day forward forecast of catalog releases.

Seven adoption specifications share the earlier full predictor backbone: ridge and boosting controls, ridge plus past release counts, ridge plus observed benchmark marks, ridge plus benchmark and NHPP predictors, a ridge model adding industry interactions with the release features, and boosting with the full release dictionary. Industry interactions include current frontier, capability stock, expected arrivals, and expected frontier gain; a frontier-by-current-adoption interaction is also included. Regularization and boosting complexity are selected using the same nested temporal folds as their controls. Generated predictors are precomputed at each historical issue date without substituting realized later arrivals. Point predictions use generated feature means; they are not a joint likelihood fit of adoption and release arrivals, and do not constitute calibrated adoption prediction intervals.

4 Results

4.1 Forecast accuracy and industry heterogeneity

Table 1 reports all forecast specifications on a common evaluation sample. Separate industry models have the lowest aggregate MAE, 3.04 pp, compared with 3.39 for the six-release mean and 3.30 for the original pooled baseline. These differences correspond to reductions of 10.3% and 8.0%, respectively. Elastic net (3.07), gradient boosting (3.12), and ridge (3.13) also improve on the recent-average benchmark. Partial pooling yields 3.29 pp, close to the baseline. The full unregularized dictionary yields MAE of 5.68 pp and produces six predictions outside the percentage range.

Table 1: Forecast errors in percentage points. Historical: 635 forecasts at nine origins; extension: 71 forecasts at one origin; all: 706 forecasts.

Model	Historical MAE	Extension MAE	All MAE	All RMSE	All bias
Persistence	4.18	3.90	4.15	5.75	-1.05
Six-release mean	3.40	3.34	3.39	4.64	-2.07
Baseline OLS	3.33	3.09	3.30	4.55	+0.07
Linear subset	3.39	3.06	3.36	4.63	+0.03
Nonlinear subset	3.36	2.97	3.32	4.66	+0.12
Full OLS	5.92	3.52	5.68	7.27	+2.11
Ridge	3.15	2.99	3.13	4.32	+0.47
Elastic net	3.08	2.93	3.07	4.29	+0.06
Gradient boosting	3.12	3.09	3.12	4.38	-0.03
Partial pooling	3.33	2.97	3.29	4.51	+0.12
Industry models	3.04	3.02	3.04	4.29	-0.21

Industry models beat the recent mean at nine of ten issue dates. Their mean origin-level

difference is −0.35 pp, with exploratory block interval [−0.46, −0.24]. Relative to baseline OLS, the corresponding difference is −0.26 pp, with interval [−0.40, −0.12]. Ridge and boosting beat the recent mean at all ten dates. These results support industry heterogeneity and regularization within this sample, although the short evaluation history limits claims about future performance.

Table 2: Paired origin-level MAE differences relative to the six-release mean. Negative values favor the fitted model. Intervals are 95% moving-block bootstrap intervals and are exploratory.

Model	Difference (pp)	Interval (pp)	Origins won
Baseline OLS	-0.09	[-0.22, +0.04]	7/10
Linear subset	-0.03	[-0.15, +0.10]	4/10
Nonlinear subset	-0.07	[-0.20, +0.08]	6/10
Full OLS	+2.29	[+0.71, +3.91]	1/10
Ridge	-0.25	[-0.36, -0.17]	10/10
Elastic net	-0.32	[-0.43, -0.21]	9/10
Gradient boosting	-0.27	[-0.45, -0.13]	10/10
Partial pooling	-0.10	[-0.23, +0.05]	5/10
Industry models	-0.35	[-0.46, -0.24]	9/10

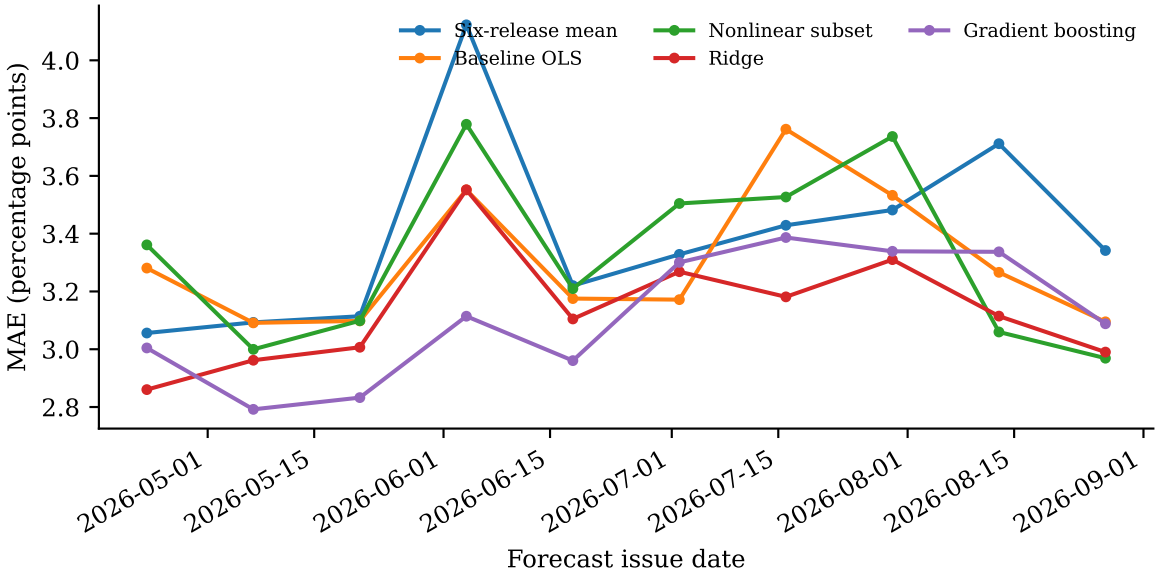


Figure 1: Forecast MAE by issue date. All lines use identical evaluated group-release pairs. The final point forecasts the extension outcome.

Industry models reduce MAE for the 50–99 employee band from 5.32 pp under the baseline to 4.23 pp, below the recent mean's 4.65 pp. The advantage is not uniform: for the 10–19 employee band, ridge and elastic net perform better than separate industry models. At the industry level, Information remains difficult: its industry-model MAE is 6.48 pp. Aggregate improvement therefore does not justify equally confident forecasts for every group. The median published target standard error is 2.83 pp; 88.8% of industry-model errors fall within twice the corresponding target standard error. This comparison describes the scale of survey uncertainty and does not establish interval calibration.

The extension contains one origin and cannot establish general superiority. Elastic net has its lowest extension MAE at 2.93 pp, followed by nonlinear subset selection at 2.97 pp. Industry models achieve 3.02 pp, compared with 3.09 for the baseline and 3.34 for the recent mean. All expanded methods remain retrospectively specified.

4.2 Selected nonlinear structure

The nonlinear search selects an industry-by-current-use interaction at every outer origin. Expected use squared and the current-use-by-time interaction are selected at six of ten origins; the current-by-expected-use interaction appears at the other four. Current use squared and time squared are never selected. Thus the candidate search finds industry-dependent slopes and some curvature in expectations, without evidence that a quadratic current-use response is consistently needed.

The nonlinear subset achieves aggregate MAE of 3.32 pp, compared with 3.36 for the linear subset and 3.30 for the baseline. Its mean origin-level improvement over the linear subset is only 0.03 pp, with interval $[-0.068, 0.005]$ for the nonlinear-minus-linear difference. Relative to the baseline, its difference is +0.02 pp with interval $[-0.064, 0.120]$. These comparisons provide limited support for an incremental predictive benefit from the nonlinear terms.

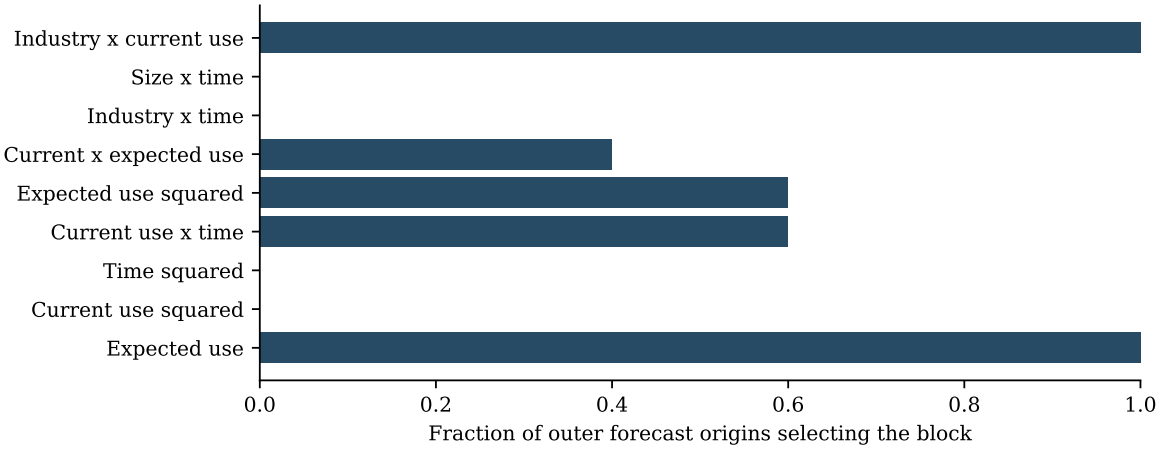


Figure 2: Selection frequency of optional blocks. Main effects for current use, time, industry, and size are mandatory. An interaction with industry allows slopes to differ across sectors.

Ridge ablations reinforce the importance of recent history. Removing lagged rates, recent change, and the rolling mean increases MAE from 3.13 to 3.70 pp. Removing the six-release cycle changes MAE only to 3.15 pp. Removing quantitative curvature terms gives 3.12 pp, and omitting contemporaneous aggregate rates also gives 3.12 pp. These small latter differences do not establish useful seasonality or a material gain from the larger dictionary. The ridge search selects penalty 1,000 at all origins, even after expanding the grid to 100,000.

4.3 Industry growth and ranking

Finance and insurance has the highest estimated growth, 1.28 pp per four weeks, followed by Information at 1.22 and professional and technical services at 1.08. Their conditional intervals are $[1.07, 1.48]$, $[0.95, 1.50]$, and $[0.94, 1.21]$, respectively. Accommodation and food services grows by 0.16 pp per four weeks, with interval $[0.11, 0.21]$. All retained industries have positive fitted

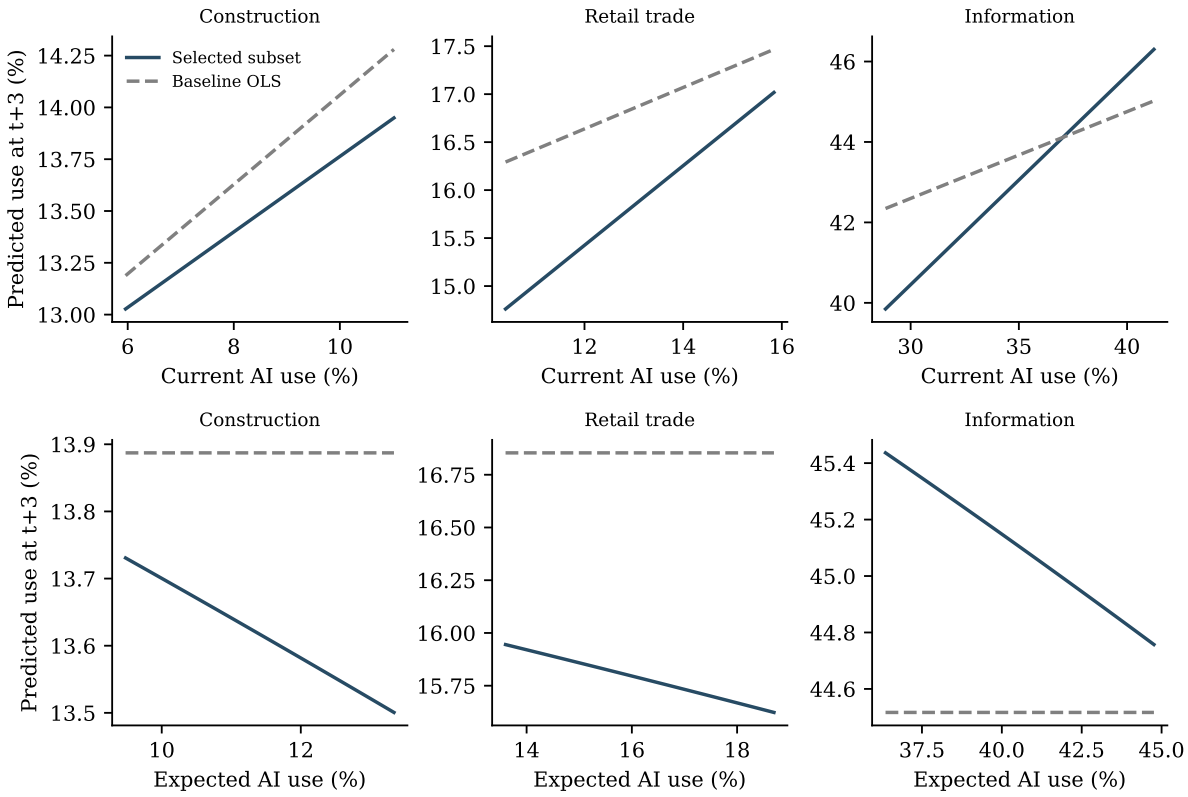


Figure 3: Conditional predictions from the final nonlinear subset and baseline, for 1–4 employee firms in three industries. The varying predictor spans its within-group training 5th–95th percentiles. The other adoption predictor is fixed at its within-group training median, and time is fixed at the final issue date. The final subset includes expected use squared; its current-use relationship is linear when time and expectations are held fixed. These curves describe fitted relationships without causal interpretation.

trends in the observed period. These are absolute percentage-point changes; rankings by relative percentage growth would answer a different question.

The top two industries cannot be confidently ordered. The Information-minus-finance growth difference is -0.06 pp per four weeks, with interval $[-0.38, 0.27]$. Finance has a bootstrap rank range of 1–3, Information 1–4, and professional services 2–3. Finance, Information, and professional services occupy the top three in 100.0%, 97.2%, and 97.8% of bootstrap draws, respectively. These frequencies are conditional on the fitted trend model and a short series. They support a leading group more strongly than a precise ordering within that group.

Adding a common sample-change indicator preserves the primary ordering closely. Inverse-variance weighting changes the relative ordering of the top two: Information's estimated growth becomes 1.16 pp per four weeks and finance's 1.14. This sensitivity is consistent with their overlapping uncertainty. The leading group remains substantively similar.

4.4 Release chronology

The event analysis uses 65 groups with observed rates at all 23 releases. The three earliest announcements have fewer than three pre-event reference periods in the comparable series and cannot

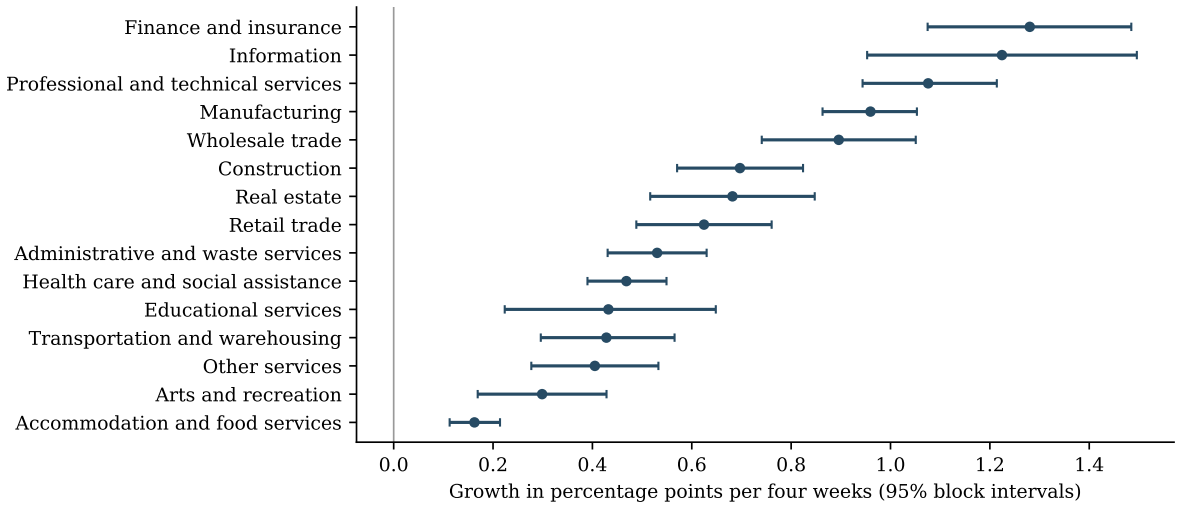


Figure 4: Industry-specific growth with conditional 95% wild block bootstrap intervals. Group intercepts account for stable industry-size differences. Results apply to retained size groups and are not population-weighted industry estimates.

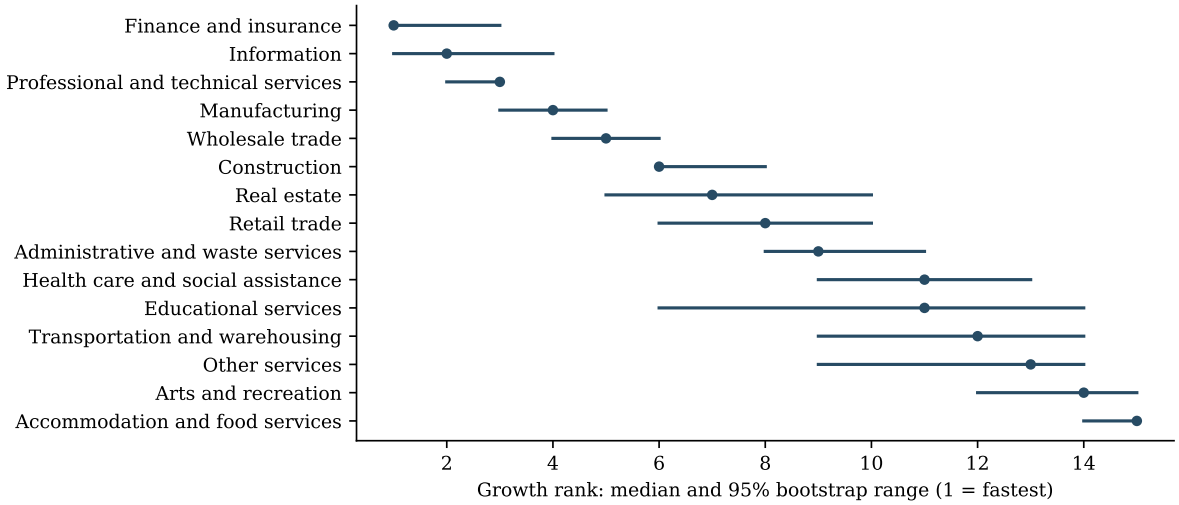


Figure 5: Median growth ranks and central 95% bootstrap rank ranges. Overlapping ranges indicate uncertainty in exact ordering; the ranges are not simultaneous confidence sets.

support the specified window comparison. For the February and March events with sufficient observations, three-period pre/post changes are positive: 0.45 pp for Claude Opus 4.6, 1.43 pp for Gemini 3.1 Pro, and 1.61 pp for GPT-5.4. The reference windows overlap substantially.

The cumulative announcement count correlates 0.81 with time. After adjustment for a linear trend and the sample-change indicator, its descriptive coefficient is -0.47 pp per selected announcement. This sign differs from the positive unadjusted window changes. The contrast illustrates sensitivity to trend adjustment and the limited independent information in a common event chronology. It does not establish a negative effect of model releases. The data do not separate release effects from ongoing diffusion or other calendar-time influences.

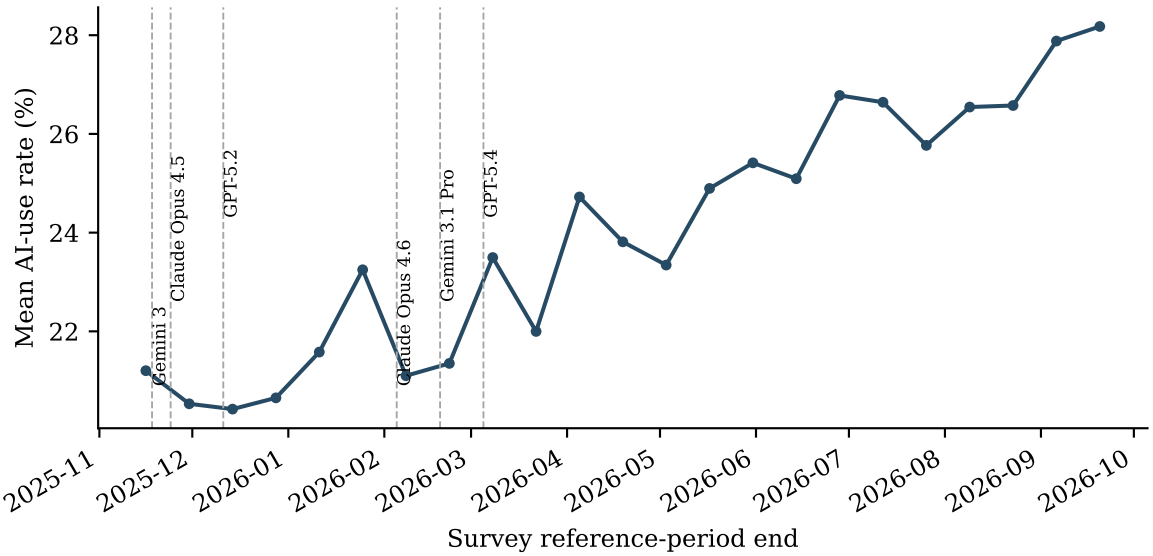


Figure 6: Balanced-group AI-use mean by reference-period end, with selected provider announcement dates. Publication occurs after the reference period and collection. Event positions indicate temporal correspondence; the figure does not estimate causal effects.

4.5 Interpolation and overparameterization

The random-feature design reaches full row rank, including the intercept, near $p/n = 1$. Minimum-norm training error approaches zero there. Median future RMSE rises from 5.63 pp at $p/n = 0.2$ to approximately 4,577 pp at interpolation. It then falls to approximately 190 pp at $p/n = 1.2$, remaining near 203 pp at $p/n = 2$. This post-peak reduction is a second-descent pattern over part of the tested range, but its prediction errors remain far above those of the practical forecasting models.

Ridge avoids interpolation and keeps median future RMSE near 4.77–4.89 pp for p/n between 1 and 2. At $p/n = 2$, its median effective degrees of freedom are approximately 210, compared with a median training-row count of 912. Thus a nominally overparameterized representation can remain strongly regularized. This experiment establishes the existence of interpolating fits in the chosen representation, while providing no evidence that unregularized overparameterization improves useful forecasting performance.

The practical full OLS model already performs poorly with 81 nominal coefficients and training samples of 560–1,195 rows. Its recorded ranks are 78–80 because the dictionary includes redundant relationships. Predictive instability therefore arises well before the formal $p > n$ regime.

5 Benchmark-Marked Release Forecasts

5.1 Benchmark evidence and estimated arrival rate

The catalog contains 271 scored model records, including 221 releases in the intensity-estimation window. Table 3 attaches benchmark scores to the six announcements considered in the earlier event chronology; the full catalog supplies the rate model. A selected six-event timeline would not provide an adequate denominator for estimating the broader release rate.

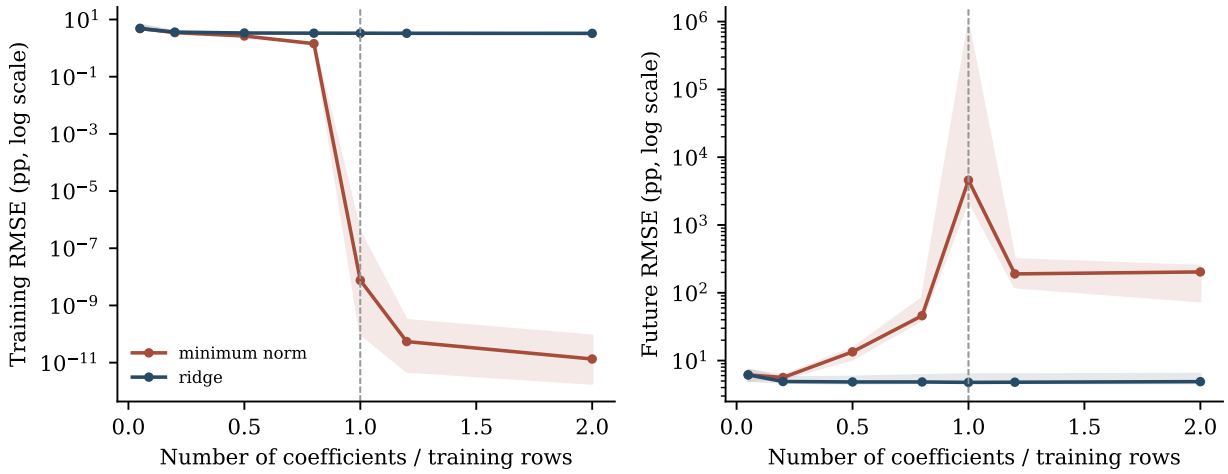


Figure 7: Training and future RMSE across model widths, using logarithmic error axes. Lines show medians over three origins and three feature seeds; shading shows the 10th–90th percentiles across those nine fits, rather than confidence intervals. The vertical line marks $p/n = 1$. Ridge uses a fixed diagnostic penalty of 0.01.

Table 3: Benchmark marks for the six announcements in the earlier chronology. ECI and its 90% intervals use the October 30 Epoch vintage. GPQA Diamond is a percentage, summarized by the within-model median across listed evaluation variants; missing scores are not inferred.

Canonical model	Release date	ECI	ECI interval	GPQA (%)
Gemini 3 Pro	2025-11-18	153.22	[151.08, 155.57]	93.9
Claude Opus 4.5	2025-11-24	150.20	[148.04, 153.23]	84.8
GPT-5.2	2025-12-11	153.83	[152.19, 155.93]	92.2
Claude Opus 4.6	2026-02-05	155.85	[154.09, 157.98]	91.0
Gemini 3.1 Pro	2026-02-19	155.24	[153.31, 157.73]	96.5
GPT-5.4	2026-03-05	157.14	[155.20, 159.47]	94.9

The rolling rate comparison selects linear time plus weekday. Its estimated intensity is

$$\widehat{\lambda}_d = \exp(-4.246 + 0.237 t_d + 2.656 W_d) \quad \text{catalog releases per day.} \tag{9}$$

The time coefficient is approximately 0.237 per year, with HAC interval $[0.008, 0.467]$. Holding weekday status fixed, the annual rate multiplier is approximately 1.27. This describes the scored catalog's release rate and may reflect changes in which models receive benchmark coverage. It does not estimate a population-wide rate of all LLM introductions.

The selected model's mean daily validation NLL is 0.671, compared with 0.759 for constant intensity. Its 42-day count MAE is 3.39, compared with 4.75 for constant intensity. The time-only model has count MAE of 3.38 but a worse daily log score, illustrating the difference between predicting total counts and their calendar distribution. All 23 survey issue dates independently select the time-plus-weekday specification using earlier release history.

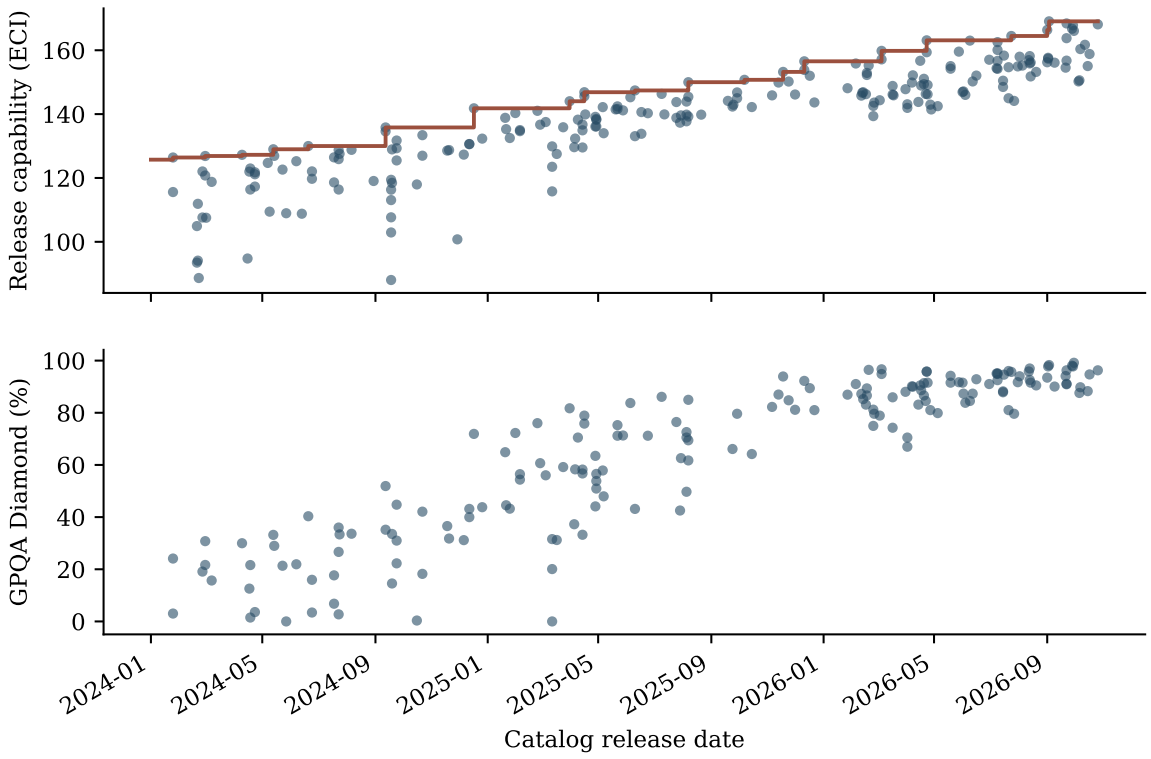


Figure 8: Release capability marks and the running ECI frontier, with GPQA Diamond scores shown separately. Scores use the October 30, 2026 vintage and are not historical score-publication snapshots. The plotted GPQA values are model-level medians across listed variants.

Table 4: Release-rate model selection using six earlier, disjoint 42-day validation windows. NLL is mean daily negative log likelihood; dispersion uses the final fit. Forecast counts cover October 31–December 11, 2026.

Rate regression	Validation NLL	Count MAE	Expected count	Dispersion
Constant	0.759	4.75	8.98	1.55
Linear time	0.744	3.38	12.51	1.59
Quadratic time	0.751	3.69	14.45	1.59
Time + weekday	0.671	3.39	12.48	1.37
Time + annual cycle	0.742	3.72	9.06	1.60

5.2 Future arrival and capability forecasts

For October 31 through December 11, 2026, the selected rate regression gives

$$\widehat{\Lambda}_{42} = \sum_{d=\text{Oct. } 31}^{\text{Dec. } 11} \widehat{\lambda}_d = 12.48. \tag{10}$$

Across 2,000 conditional simulations with parameter uncertainty, the central 90% count interval is 6–20 catalog releases. A moment-matched overdispersion sensitivity expands it to 6–21. These are

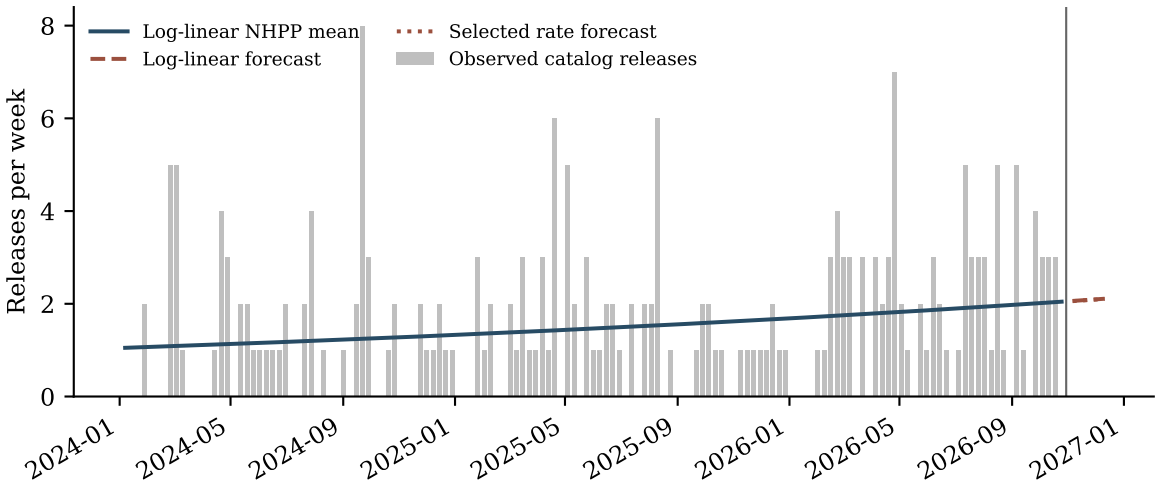


Figure 9: Observed weekly catalog arrivals and the fitted log-linear NHPP mean. Dashed forecast curves show the time-only and selected time-plus-weekday regressions. Forecast points sum daily intensities over six complete, nonoverlapping seven-day intervals beginning October 31. The incomplete historical calendar week is omitted. The vertical line marks October 30, 2026.

model-based predictive ranges for this catalog definition, rather than a census forecast of all model releases.

The mark regression estimates an average ECI slope of 17.40 points per year among scored releases, with provider/week-clustered standard error 1.05. Its residual standard deviation is 7.65 ECI points. Starting from the observed frontier of 169.06, simulated frontier ECI after 42 days has mean 172.39 and central 90% interval $[169.06, 176.44]$. The lower endpoint reflects the possibility that no arriving model exceeds the current frontier. These projections depend on extrapolating the release-mark distribution; they are not predictions of particular models or independently calibrated frontier intervals.

5.3 Poisson assumptions and sensitivity

The selected rate fit has Pearson dispersion 1.37, above the Poisson value of one, and residual lag-one correlation of 0.073. There are 47 days with at least two catalog releases, compared with approximately 26.5 expected under the selected independent-arrival model. None of 2,000 plug-in Poisson replications has as many such days. This is a model diagnostic, rather than a formally calibrated test following model selection. The evidence indicates residual batching beyond weekday effects.

Collapsing releases to provider-day batches reduces the intensity sample from 221 model events to 179 batch events. In the time-only regression, dispersion falls from 1.59 to 1.10. Its 42-day forecast is 11.45 provider-day batches, which has a different unit from the model-count forecast. The sensitivity supports joint-announcement behavior as one explanation for overdispersion. A Poisson rate regression remains useful as a conditional mean specification, while its independent-increment predictive assumptions require qualification.

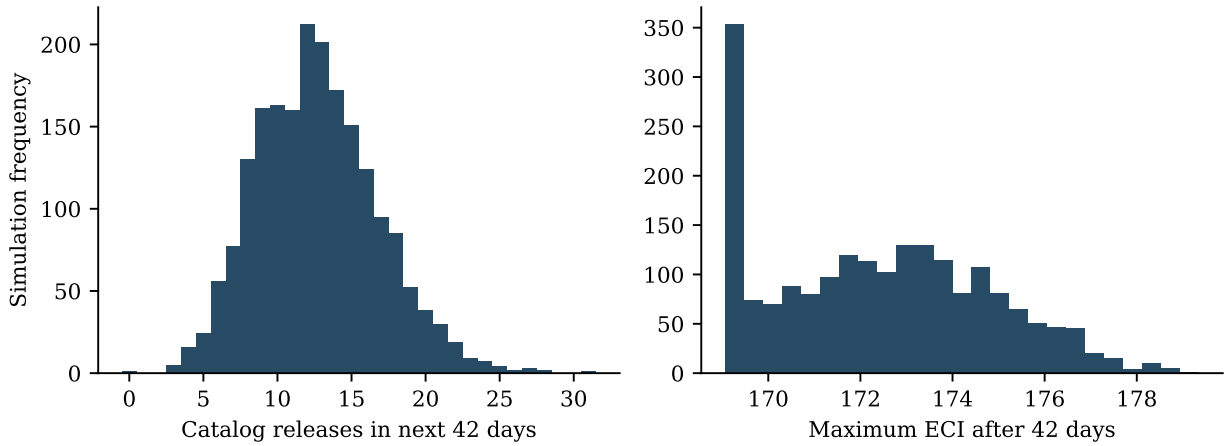


Figure 10: Conditional simulations of 42-day catalog release counts and maximum ECI. Rate and mark coefficient uncertainty is included approximately. The frontier distribution has mass at the existing maximum. Benchmark-calibration uncertainty and future catalog-selection changes are not fully represented.

5.4 Incremental value for adoption forecasting

Table 5 evaluates release-augmented regressions on the original 706 forecasts. The matched ridge control reproduces the earlier ridge MAE of 3.135 pp. Adding observed release counts gives 3.202 pp, observed benchmark features give 3.187 pp, and benchmark plus NHPP features give 3.267 pp. Industry interactions reduce the latter to 3.241 pp. The extension-date results are more favorable for some augmented models, but represent only one forecast origin.

Table 5: Adoption forecast errors in percentage points on the same 706 forecasts. Controls use the same feature backbone, tuning grids, and earlier validation dates as their augmented counterparts.

Specification	All MAE	All RMSE	Historical MAE	Extension MAE
Ridge control	3.135	4.323	3.151	2.990
Boosting control	3.115	4.365	3.119	3.079
Counts + ridge	3.202	4.398	3.180	3.394
Benchmarks + ridge	3.187	4.353	3.209	2.988
Benchmarks + NHPP + ridge	3.267	4.431	3.301	2.962
NHPP + industry interactions	3.241	4.405	3.276	2.934
Benchmarks + NHPP + boosting	3.120	4.360	3.125	3.078

Matched boosting controls provide a separate comparison for the nonlinear learner. Boosting without release features has MAE of 3.115 pp, compared with 3.120 pp when benchmark and NHPP features are added. The origin-level difference is +0.004 pp, with exploratory block interval $[-0.024, 0.033]$. Hence a low overall error from the augmented boosting model cannot be attributed to the release features. The earlier separate-industry model remains more accurate overall, at 3.039 pp.

Table 6: Paired origin-level MAE differences. Negative values favor the first specification. Intervals are exploratory 95% moving-block bootstrap intervals over ten dates.

Comparison	Difference (pp)	Interval (pp)	Origins won
Benchmark ridge vs. Ridge control	+0.052	[-0.052, +0.179]	5/10
NHPP ridge vs. Ridge control	+0.132	[-0.023, +0.358]	5/10
NHPP ridge vs. Benchmark ridge	+0.080	[-0.011, +0.189]	5/10
Interactions vs. NHPP ridge	-0.026	[-0.074, +0.011]	7/10
NHPP boosting vs. Boosting control	+0.004	[-0.024, +0.033]	6/10

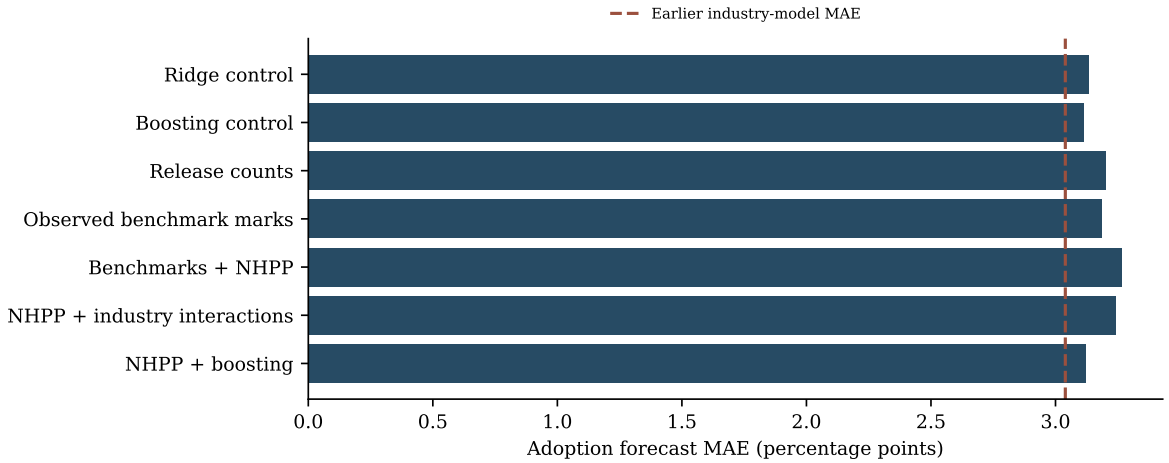


Figure 11: Adoption MAE for controls and release-augmented specifications. All models use identical outer forecast observations and nested temporal tuning. The dashed line is the earlier separate-industry model's MAE. The experiment is retrospective because historical benchmark score vintages are unavailable.

The additional regressions satisfy the proposed stochastic-release mechanism, but the available sample does not establish incremental adoption-prediction value. Benchmark features are shared across groups at a date and correlate with calendar time, while survey rates have substantial measurement noise. With only ten evaluation origins, these factors limit the ability to distinguish a benchmark-release mechanism from general adoption trends. No causal effect of release frequency or benchmark performance is identified.

6 Discussion

The clearest forecasting result is the value of allowing industries to differ. Separate industry regressions reduce average error relative to the original pooled specification and improve the previously weak 50–99 employee band. Regularized pooled models also improve performance. The unrestricted full dictionary performs substantially worse, and removing quantitative curvature from ridge has little effect. These findings favor targeted heterogeneity, historical smoothing, and coefficient control over indiscriminate expansion of the regression equation.

Nonlinear subset selection identifies industry-dependent slopes and a curved association between expectations and future published use. The selected expectation form changes across validation dates and does not deliver an aggregate improvement over the baseline. The final subset gives a

decreasing conditional relationship with expected use over the plotted ranges, further limiting a structural interpretation. The statistical conclusion is therefore limited: the candidate dictionary contains nonlinear descriptions, but their incremental predictive value has not been established. A subsequent preregistered comparison on genuinely future releases would provide stronger evidence.

For industry comparisons, finance, Information, and professional services form a relatively stable leading growth group. Their exact ordering is uncertain. Outreach organizations could use these results to distinguish broad adoption trajectories while avoiding precise league tables. Higher adoption or faster growth does not establish unmet need, accessibility of assistance, or the benefit of an intervention. Resource-allocation decisions require additional information on barriers, costs, and program outcomes.

The interpretation of uncertainty remains conditional. Estimates arise from a short, noisy panel with recurring groups, shared date shocks, and a rotating business sample. The six-release block bootstrap has few effective temporal blocks. It is not a substitute for Census replicate weights or full design-based covariance information. Standard errors address sampling variability, while nonresponse and interpretation changes can introduce additional error. Screening excludes groups with frequent missing estimates, and the results may not transfer to those groups. Equal weighting across retained cells also differs from weighting industries by their business populations.

The event study supplies a useful chronology but cannot attribute adoption changes to specific releases. Announcements are common exposures, cluster early in the sample, and overlap with a strong adoption trend. Early releases lack comparable pre-event data. A stronger design would link firm or industry exposure to particular tools, specify comparison groups before inspecting outcomes, and assess pre-event trends over a longer consistent series.

The benchmark-marked release model replaces a selected announcement count with an explicit stochastic arrival mechanism and a regression for capability marks. Its time-plus-weekday intensity improves the release-model selection criterion over a constant rate, but residual batching weakens independent-Poisson assumptions. The resulting future-arrival and capability features show no clear incremental adoption accuracy over matched controls. This result distinguishes successful construction of a release forecast from evidence that the forecast is useful for predicting business adoption. Historical score-publication snapshots, more independent survey dates, and exposure measures linking businesses to particular models would strengthen a subsequent test.

Finally, the capacity experiment shows why parameter count alone is an inadequate guide to forecasting. Minimum-norm interpolation produces extreme errors in this noisy setting, although error declines after the interpolation peak. Ridge stabilizes a much larger feature representation through shrinkage. These results are specific to the tested random-feature distribution, fixed bandwidth, penalty, origins, and seeds; they do not establish a universal double-descent curve for BTOS or for machine learning generally.

7 Conclusion

Industry-specific regression reduces aggregate MAE by 10.3% relative to a six-release mean in this recorded sample. Elastic net and gradient boosting also improve accuracy. Nonlinear subset selection yields unstable functional forms and no aggregate improvement over the baseline. Overparameterized minimum-norm fits interpolate the training data and produce excessive future errors; regularization stabilizes this experiment. The fitted nonstationary release regression forecasts benchmarked-catalog arrivals, while its independent-Poisson assumptions are weakened by batching. Benchmark scores and expected arrivals do not show a clear adoption-forecast gain over matched controls in the available survey window. The practical value of a forecast depends on

stable performance across issue dates and relevant business groups. Ranking uncertainty should accompany industry comparisons. Release timing provides hypotheses for further investigation, while credible causal attribution would require additional exposure information or an appropriate comparison design.

Acknowledgments

AI assistance was used to develop the analysis code and revise the manuscript. The original Milestone 2 report and scripts are preserved in the repository.

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A Additional results and reproducibility

Table 7: Industry growth in percentage points per four weeks. Intervals and rank ranges summarize the conditional wild block bootstrap; top-three shares are resampling frequencies.

Industry	Growth	95% interval	Rank range	Top three
Finance and insurance	1.28	[1.07, 1.48]	1–3	100.0%
Information	1.22	[0.95, 1.50]	1–4	97.2%
Professional and technical services	1.08	[0.94, 1.21]	2–3	97.8%
Manufacturing	0.96	[0.86, 1.05]	3–5	3.8%
Wholesale trade	0.90	[0.74, 1.05]	4–6	1.3%
Construction	0.70	[0.57, 0.82]	6–8	0.0%
Real estate	0.68	[0.52, 0.85]	5–10	0.0%
Retail trade	0.62	[0.49, 0.76]	6–10	0.0%
Administrative and waste services	0.53	[0.43, 0.63]	8–11	0.0%
Health care and social assistance	0.47	[0.39, 0.55]	9–13	0.0%
Educational services	0.43	[0.22, 0.65]	6–14	0.0%
Transportation and warehousing	0.43	[0.30, 0.57]	9–14	0.0%
Other services	0.40	[0.28, 0.53]	9–14	0.0%
Arts and recreation	0.30	[0.17, 0.43]	12–15	0.0%
Accommodation and food services	0.16	[0.11, 0.21]	14–15	0.0%

Table 8: Sensitivity of growth estimates (pp per four weeks) to a common sample-change indicator and inverse-variance weighting. Primary confidence intervals do not apply to these sensitivity fits.

Industry	Primary	Sample adjusted	SE weighted
Finance and insurance	1.28	1.29	1.14
Information	1.22	1.24	1.16
Professional and technical services	1.08	1.09	0.96
Manufacturing	0.96	0.97	0.87
Wholesale trade	0.90	0.91	0.76
Construction	0.70	0.71	0.72
Real estate	0.68	0.70	0.65
Retail trade	0.62	0.64	0.45
Administrative and waste services	0.53	0.54	0.52
Health care and social assistance	0.47	0.48	0.55
Educational services	0.43	0.45	0.50
Transportation and warehousing	0.43	0.44	0.40
Other services	0.40	0.42	0.36
Arts and recreation	0.30	0.31	0.15
Accommodation and food services	0.16	0.18	0.17

Table 9: Nonlinear-subset selection frequency over ten outer origins. Baseline main effects are mandatory.

Optional block	Selected origins	Frequency
Expected use	10	100%
Current use squared (y_t^2)	0	0%
Time squared (t^2)	0	0%
Current use × time	6	60%
Expected use squared (e_t^2)	6	60%
Current × expected use	4	40%
Industry × time	0	0%
Size × time	0	0%
Industry × current use	10	100%

Table 10: MAE by employment-size band over all ten evaluation origins.

Employees	Six-release mean	Baseline	Industry	Elastic net	Boosting
1–4	2.53	2.10	1.92	1.87	1.75
5–9	3.15	3.32	3.01	2.98	3.17
10–19	3.41	2.84	2.96	2.68	2.78
20–49	3.56	3.50	3.41	3.58	3.61
50–99	4.65	5.32	4.23	4.69	4.72

Table 11: Ridge feature ablations, each tuned independently using the same inner temporal protocol.

Dictionary	MAE (pp)	RMSE (pp)
full	3.135	4.323
no aggregates	3.117	4.382
no curvature	3.117	4.257
no cycle	3.147	4.348
no history	3.701	5.009

Table 12: Changes in the balanced-group mean from three pre-announcement reference periods to three wholly subsequent periods. These overlapping comparisons are descriptive.

Announcement	Date	Pre	Post	Change (pp)
Gemini 3	2025-11-18	1	3	Insufficient history
Claude Opus 4.5	2025-11-24	1	3	Insufficient history
GPT-5.2	2025-12-11	2	3	Insufficient history
Claude Opus 4.6	2026-02-05	3	3	+0.45
Gemini 3.1 Pro	2026-02-19	3	3	+1.43
GPT-5.4	2026-03-05	3	3	+1.61

Table 13: Median errors over three origins and three feature seeds. MN denotes minimum-norm least squares; degrees of freedom (df) are for ridge. Each origin has its own training-row count; medians are descriptive summaries.

p/n	MN train RMSE	MN test RMSE	Ridge train	Ridge test	Ridge df
0.05	4.9	6.16	4.86	6.15	45.8
0.20	3.5	5.63	3.62	4.91	141.1
0.50	2.7	13.46	3.37	4.84	184.4
0.80	1.4	45.97	3.33	4.84	196.2
1.00	7.6e-09	4576.79	3.31	4.77	199.7
1.20	5.5e-11	189.86	3.29	4.80	202.8
2.00	1.3e-11	202.99	3.27	4.89	209.7

A.1 Computational record

The repository includes the unchanged original report and code, an attributed source extract and hash, all predictions, selected blocks, tuning parameters, validation audits, industry bootstrap draws, event windows, complexity diagnostics, and the LaTeX source. Reproduction requires Python with the packages listed in `requirements.txt`. The commands are:

```
python -m unittest test_analysis test_release_process -v
python extended_analysis.py --workbook data/btos_ai_extract_2026-10-15.csv
python release_process.py --as-of 2026-10-30 --simulations 2000
python validate_release_results.py
python build_report.py
cd report
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

Regression tests verify the original counts and rounded performance values, temporal training boundaries, hierarchy constraints, training-only preprocessing, rank handling, and recovery of a quadratic signal in synthetic data. Release-process checks verify zero-event days, date filtering, integrated intensities, simulation ranges, and matched adoption controls. Benchmark snapshots and provenance hashes are committed. The original baseline has 21 coefficients. Full-model column counts and ranks are recorded separately; redundant columns do not count as independent information.